

Praveen Kumar

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Location: Hyderabad

Professional Summary

Highly motivated Machine Vision and Embedded AI engineer with experience in developing real-time inspection and monitoring systems, seeking to leverage my skills in AI and Python to contribute to the development of innovative Conversational AI solutions as an Agent Designer at PolyAI. With a strong foundation in machine learning, computer vision, and edge AI, I am excited to explore the possibilities of LLMs and Conversational AI at PolyAI.

Technical Skills

- **Programming:** Python, Java, SQL
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, Keras
- **Computer Vision:** YOLOv8, OpenCV, Object Detection, Image Processing
- **Edge AI:** Raspberry Pi, Embedded Inference, Real-Time Processing
- **IoT & Embedded:** Arduino, Sensor Integration, Control Systems
- **Sensors:** LiDAR, Ultrasonic, Thermal, Optical, Gyroscope
- **Tools:** Git, Jupyter Notebook, Twilio API, CAD

Education

Bachelor of Technology in Computer Science (AI & ML), Woxsen University, Hyderabad, 2022 – 2026

Key Projects

- **Railway AI Research (Patent Published):** Designed AI-enabled wheel inspection system using machine vision and sensor fusion, integrated LiDAR, thermal, ultrasonic, and optical sensors for defect detection, and developed microcontroller-based real-time diagnostic and IoT monitoring system.
- **Smart Crop Guard – Real-Time Edge AI Monitoring System:** Developed embedded machine vision system using YOLOv8, built full AI pipeline, integrated sensors, lights, and speakers with automated alerts via Twilio API, and designed temporal filtering logic reducing false detections.
- **AI-Powered Smart Irrigation System:** Developed EfficientNet-B0 model, built real-time inference pipeline on Raspberry Pi with camera and soil sensors, and designed offline irrigation decision logic

using vision and sensor data.

- **Patented Dual-Axis Solar Tracking System:** Designed and patented dual-axis solar tracking system, developed motor control algorithms, built embedded control system integrating sensors and microcontroller-based tracking logic, and developed TRL-4 hardware prototype.
- **Smart Waste Segregation System:** Designed automated waste segregation system, developed embedded control logic, built rotating separation mechanism, and reduced manual sorting effort by 35%.

Achievements & Publications

- **IEEE Publication** – Smart Crop Guard: AI-Enabled Wildlife and Adaptive Deterrence System
- **Patent Granted** – Sun Tracking Solar Panel System
- **Patent Published** – Wheel Condition Assessment Device (Application No:-202541067091)
- **Co-author** – IoT Applications Book Chapter Role of IoT In Communications (Taylor & Francis)
- **Treasurer** – IEEE Student Branch
- **Shortlisted** – MSME Zonal SIH 2024